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Alfvén's critical ionization velocity observed in high power impulse magnetron sputtering discharges

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Azimuthally rotating dense plasma structures, spokes, have recently been detected in several high power impulse magnetron sputtering (HiPIMS) devices used for thin film deposition and surface treatment, and are thought to be important for plasma buildup, energizing of electrons, as well as cross-B transport of charged particles. In this work, the drift velocities of these spokes are shown to be strongly correlated with the critical ionization velocity, CIV, proposed by Alfvén. It is proposed as the most promising approach in combining the CIV and HiPIMS research fields is to focus on the role of spokes in the process of electron energization. © 2012 American Institute of Physics. [<http://dx.doi.org/10.1063/1.4752073>]

I. INTRODUCTION

The critical velocity, also called the critical ionization velocity (CIV) hypothesis by Alfvén,^{1,2} postulates that a strong increase in the ionization rate should occur if a plasma and a neutral gas component are in relative motion across a magnetic field, with a relative velocity component $v_{0\perp}$ across the magnetic field that exceeds the critical ionization velocity v_c ,

$$v_{0\perp} > v_c = \sqrt{\frac{2eU_i}{m_n}}. \quad (1)$$

Here U_i is the ionization potential and m_n is the mass of the neutral atom or molecule (the case of multiple gas and ion species is treated in Sec. III). In an $\mathbf{E} \times \mathbf{B}$ discharge, such a steep increase of the ionization, as function of the velocity, was further proposed³ to regulate the discharge to a state in which the plasma Hall drift speed in the laboratory rest frame (in which the neutral gas is assumed to be at rest) is $v_{Hall} = E/B \approx v_c$. Alfvén's original line of argument, also most usual in the CIV literature, is based on the following energy budget analysis for the electron population, made in the rest frame of the plasma. One electron impact ionization costs the electrons the energy eU_i , but also creates a new ion which starts with the initial velocity $-v_{0\perp}$ relative to the plasma. This ion carries a kinetic energy $W_{kin} = m_n v_{0\perp}^2/2$. If this energy can be transferred to the electrons, and the whole process is 100% effective, a positive feedback loop becomes energetically possible when $W_{kin} \geq eU_i$. This condition directly gives Eq. (1). Including efficiencies that are below 100% leads to a lower effective critical ionization velocity, commonly expressed in the form $v_c^* = v_c/\sqrt{\eta}$, where the constant η always is smaller than unity. η was originally proposed⁴ to represent the fraction of the energy W_{kin} that is transferred to the electrons but was later⁵ extended to include also other reasons for reduced efficiency, e.g., hot electron losses out of the active region, and an effective cost of ionization higher than eU_i . Notice that according to Eq. (1), v_c is determined by the neutral gas species alone, independent of,

e.g., the magnetic field strength, the discharge current, the plasma density n_e , and the neutral gas density n_n .

Experimental verification of Alfvén's CIV hypothesis came about two decades later, first from pulsed high power $\mathbf{E} \times \mathbf{B}$ discharges in various geometries, then from plasma gun impact experiments and also to some extent from ionospheric injection experiments. For reviews, see Refs. 6–9. Perhaps the most remarkable finding was that Eq. (1) was found to hold within typically 50% for all tested neutral gas and plasma combinations, for several different discharge geometries, and over orders of magnitude in gas pressure, plasma density, magnetic field strength, and current density. In a comprehensive review, Brenning⁹ concluded that only two empirical conditions seem to be needed for the CIV effect to occur. First, the magnetic field has to be so strong that the Alfvén speed is $V_A = B/\sqrt{\mu_0 n_i m_i} > 3v_c$. Second, the gas density has to be so high that a Townsend condition proposed by Haerendel,¹⁰ $\int n_n dz > v_c/k_{i,max}$, is fulfilled with a margin of about a factor of two. In this condition, $k_{i,max}$ is the maximum of the electron impact ionization rate coefficient, and the integral is taken over the extent of the interaction volume in the direction z of relative motion between plasma and neutral gas. In physical terms, the Townsend condition means that the neutral gas has to be so dense that electrons, if efficiently energized, have a chance to ionize when crossing the interaction volume. In laboratory $\mathbf{E} \times \mathbf{B}$ discharges, there is also an upper limit to the gas density at about 10 Pa (Ref. 9) above which the speed of spokes and ionizing fronts can be reduced below the critical velocity of Eq. (1). This is understood as due to gas pick-up and well described by the snow plow model.¹¹ We will here discuss experiments performed at typically 0.3 Pa, far below the range of $p > 10$ Pa, where this happens.

The present work is motivated by the recent discovery^{12–14} of rotating dense plasma structure, spokes, in high power impulse magnetron sputtering (HiPIMS) discharges. For reviews of the HiPIMS field, see, e.g., Refs. 15–17. HiPIMS discharges have many similarities with the earlier CIV experiments: they are $\mathbf{E} \times \mathbf{B}$ discharges operated in

pulsed mode with high momentary power and low duty cycle, and they satisfy the empirical CIV criteria on magnetic field strength and neutral gas density described above. Furthermore, as we will here demonstrate, also the spokes in the HiPIMS devices rotate with the CIV speed of Eq. (1). The physical implications of this, and of the role of spokes in the HiPIMS discharges, are still largely unknown. It is, however, proposed¹⁸ that the spokes affect the plasma buildup and the energizing of electrons, as well as cross-B transport of charged particles. These processes have in turn an important impact on the resulting thin film properties,¹⁹ which emphasizes the need to investigate the physics involved.

The paper is organized as follows. In Sec. II, we give a background to spokes and ionizing fronts in the CIV research field. In Sec. III, we discuss the spokes in HiPIMS magnetrons and demonstrate that, provided that the velocities of these spokes is taken as the criterion, also these belong to the CIV class of experiments. Section IV finally gives a summary and an overview.

II. SPOKES AND IONIZING FRONTS IN CRITICAL IONIZATION VELOCITY (CIV) EXPERIMENTS

The first two devices dedicated to study the CIV phenomenon were both pulsed-power $\mathbf{E} \times \mathbf{B}$ discharges with rotational symmetry, but with different magnetic topology. In homopolar devices²⁰ the magnetic field is axial, the electric field is radial, and the plasma drift is azimuthal. In coaxial plasma guns¹¹ the magnetic field is azimuthal, the electric field is radial, and the plasma drift is axial. The discharge here takes the form of a radial current sheath, or an ionizing front, that moves down the cylindrical discharge vessel. In both types of devices, the measured drift speeds are close to the proposed v_c for all gases tested. Fig. 1 shows typical data from a coaxial plasma gun experiment by Axnäs.²¹

The first observations of spokes in connection with the CIV phenomenon were made in the experiments by Angerth *et al.*³ Based on photomultiplier and magnetic probe measurements in a homopolar device, they concluded that the discharge starts as a narrow spoke confined in azimuth, and then spreads out azimuthally when a steady state with constant discharge current is reached. This spoke phenomenon was also found to occur, and was further studied, in the Bochum I and Bochum II homopolar machines.^{22–26} For pressures below approximately 10 Pa, the spoke speed was found¹¹ to be close both to the macroscopic E/B value, and to the critical velocity of the gas in question, in agreement with the findings from the ionizing fronts in the coaxial plasma gun.

In summary, in both topologies the CIV $\mathbf{E} \times \mathbf{B}$ discharges ignite, and at least initially carry a large fraction of the discharge current, in a sheath (also called spoke, or ionizing front structure) that, in a gas pressure range $p_0 < 10$ Pa, moves in the $\mathbf{E} \times \mathbf{B}$ direction with a speed that is close to the theoretical value of v_c for the gas in question from Eq. (1).

III. SPOKE SPEEDS IN HiPIMS SPUTTERING MAGNETRONS

HiPIMS is a recent addition to sputter-based methods for thin film deposition and surface treatment based on

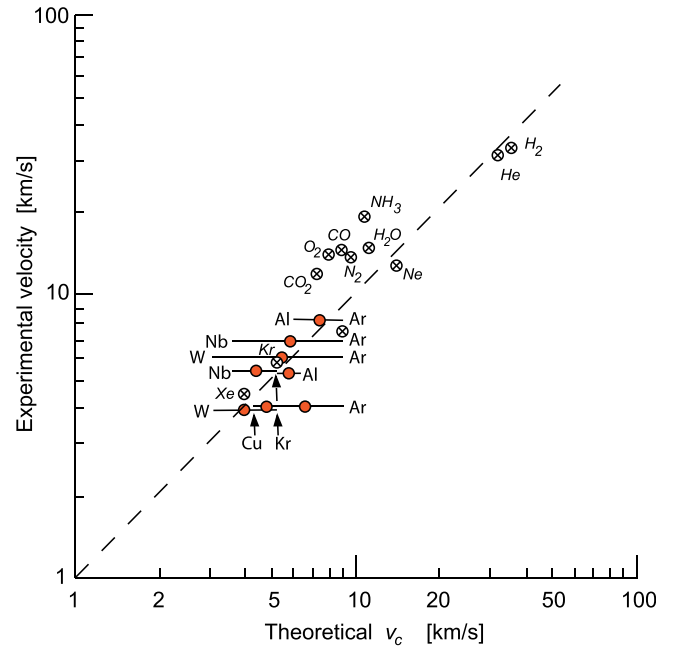


FIG. 1. An illustration of the agreement, typically within a factor of two, between the velocities of ionizing fronts, spoke velocities, and Alfvén's critical velocity hypothesis. The dashed line shows the locations in the graph where the experimental speeds would agree exactly with the theoretical v_c . The circles with crosses show data from a coaxial plasma gun CIV experiments;²¹ the $\mathbf{E} \times \mathbf{B}/B^2$ drift speed of an ionizing front moving along the axis of the device, as function of v_c for the gas species used. The horizontal bars show the spoke speeds in the HiPIMS experiment of Anders *et al.*,¹⁸ as analyzed below in Sec. III. They are located vertically at the average observed spoke speed, and range horizontally from the theoretical v_c for the pure process gas to v_c for the pure target species, as marked in the figure. To help the eye to see the trends, filled circles show the middle of each bar.

pulsed plasma discharges.^{15,16,27} In HiPIMS, a large fraction of the sputtered material is ionized, over 90% in the case of Ti,²⁸ due to a high plasma density reaching about 10^{18} – 10^{19} m⁻³ in the near-cathode region during the discharge pulse.^{29,30} Ar is commonly used as discharge gas, and a typical neutral gas density is around 10^{20} m⁻³ for a discharge pressure of 0.4 Pa. Of this, not more than about 10% of the total amount is reported to be ionized even during the most intense part of the discharge.³¹ In general, HiPIMS R&D systems are equipped with circular magnetrons resulting in $\mathbf{E} \times \mathbf{B}$ pulsed plasma discharges with rotational axial symmetry, where the electric field and the magnetic field are, respectively, axially and radially oriented above the target erosion zone (the race track). Recently, Kozyrev *et al.*,¹² Ehasarian *et al.*,¹³ and Anders *et al.*¹⁴ have independently reported the existence of azimuthally limited spoke structures in three different HiPIMS devices. Of particular interest for a comparison with the CIV experiments is that Anders *et al.* measured the azimuthal velocity of the spokes for all of the eight possible combinations of two process gases (Ar and Kr) and four target materials (Al, Cu, Nb, and W). The observed spoke velocities lay in the range 4.0 ± 0.3 km/s to 8.1 ± 0.3 km/s, where “ $\pm$ ” denotes variations of spoke velocities, not errors of measurement. As seen in Table I, these observed spoke speeds all cases except those with a Cu target lie in the range of the critical velocities for the involved species, Cu cases being close to it but not perfectly

TABLE I. Theoretical critical ionization velocities v_c for pure species, from Eq. (1), and experimental spoke velocities v_{sp} from Anders *et al.*,¹⁸ for the eight target and process gas combinations used.

Species	v_c (km/s)	v_{sp} in Kr	v_{sp} in Ar
Al	6.54	5.4 ± 0.3	8.1 ± 0.3
Cu	4.84	4.2 ± 0.15	4.2 ± 0.4
Nb	3.78	5.7 ± 0.7	7.0 ± 2.0
W	2.89	4.0 ± 0.3	6.25 ± 0.3
Ar	8.72	...	...
Kr	5.68	...	...

within it. For a more detailed analysis, the situation is complicated by the fact that the HiPIMS plasma is known, both from experiments¹⁴ and from modeling,³¹ to generally contain a mixture of process gas ions G^+ and sputtered species ions M^+ . Either one or the other can dominate depending on the pulse power and the species involved. From CIV experiments in several different binary gas mixtures, it is, however, known³² that the critical velocity is dominated by the most easily ionized component even if this represents only a minority in the neutral gas mixture. A semi-empirical extension of Eq. (1) to binary gas mixtures, which reproduces the experimentally observed trends, was proposed by Raadu³³

$$\alpha \frac{1}{2} m_{i1} v_c^2 + (1 - \alpha) \frac{1}{2} m_{i2} v_c^2 = \alpha e U_{i1} + (1 - \alpha) e U_{i2}. \quad (2)$$

Here, α is not the neutral gas mixing ratio, but the fractional production of ions of species 1, a quantity which depends both on the gas mixing ratio and the electron temperature and gives greater weight to the species with lowest ionization potential. For the unknown mixture of process gas and sputtered material in the HiPIMS device of Anders *et al.*,¹⁸ the combined critical velocity should thus lie between those of the two pure species, $v_{c,G}$ for the process gas and $v_{c,M}$ for the sputtered target (metal). The critical velocities and spoke speeds are compared in Fig. 2. The theoretical ranges between pure gas and pure target critical velocities are represented by horizontal bars. There are two combinations, Al target in Kr gas and Cu in Kr gas, where these theoretical ranges are narrow. In these cases, there is a quite good agreement with the spoke speeds: in the first case $v_{c,Al} = 6.54$, $v_{c,Kr} = 5.68$, and $v_{sp,Al/Kr} = 5.4 \pm 0.3$ km/s, and in the second case $v_{c,Kr} = 5.68$, $v_{c,Cu} = 4.84$, and $v_{sp,Cu/Kr} = 4.2 \pm 0.15$ km/s. For the remaining combinations, the uncertainty in the ion mixture leaves rather wide ranges of theoretical critical velocities. In every case, however, except for the discharges with a Cu target, these ranges span the observed spoke speed (*i.e.*, intersect the dashed line). Filled circles mark, for reference, the median critical velocity $(v_{c,M} + v_{c,G})/2$. There is a quite good agreement between these median values and the observed spoke velocities, but this must be regarded largely as a coincidence. More important are the trends with process gas changes Kr $\rightarrow$ Ar. If the higher v_c of Ar influences the result as expected from Eq. (2), the trend should be towards higher spoke velocities. This is also the case for Al ($5.4 \rightarrow 8.1$ km/s), for Nb

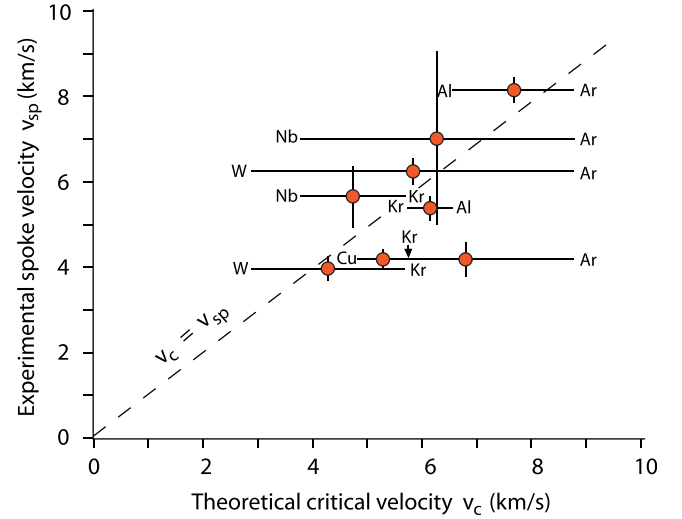


FIG. 2. A comparison between the experimentally observed spoke velocities v_{sp} from Ref. 18, and the theoretical v_c for the gases and targets used. The horizontal bars show the ranges from $v_{c,M}$ to $v_{c,G}$, according to Eq. (2), from pure target to pure process gas plasma. Filled circles mark the median values in these ranges. The vertical bars mark the observed variations of the spoke velocity. Notice that the dashed line not is a trend line for the data; it shows, for reference, the location where the identity $v_c = v_{sp}$ holds.

($5.7 \rightarrow 7.0$ km/s), and for W ($4.0 \rightarrow 6.25$ km/s). Again copper is the one exception, being the only target where the spoke speed is independent of the change in process gas. Anders *et al.*¹⁸ noted that the discharges with Cu target behave differently, and proposed this to be due to the exceptionally high self sputter yield of copper; a HiPIMS discharge with a Cu target can essentially operate in the “gas” of the sputtered metal. If the Cu/Ar discharge operates in close to pure Cu, the spoke speed should be compared to the critical velocity for pure copper, in good agreement with the data in Table I ($v_{sp} = 4.2$ km/s and $v_{c,Cu} = 4.84$ km/s). For easy cross reference, the theoretical v_c ranges and their median values, from Fig. 2, are also drawn in Figure 1 along with the ionizing front speeds in the coaxial plasma gun.²¹ Figs. 1 and 2 together show that, judging from the speeds of the spokes, the HiPIMS magnetrons belong to the class of experiments showing Alfvén’s critical velocity.

IV. SUMMARY AND DISCUSSION

There are many similarities between HiPIMS sputtering magnetron discharges on one hand, and the earlier homopolar and coaxial gun CIV experiments on the other. They are all high power pulsed $\mathbf{E} \times \mathbf{B}$ discharges, they all satisfy the empirical CIV criteria on magnetic field strength and neutral gas density described in the introduction, and they all have spokes, or fronts, with high electric currents and plasma densities, and with enhanced ionization rates. Moreover, these spokes and fronts move in the $\mathbf{E} \times \mathbf{B}$ direction with a speed typically within $\pm 50\%$ of Alfvén’s proposed critical ionization velocity of the neutral gas in question. Although this implies that there should be a lot of common ground in the physics of these two groups of discharges, one should also expect significant differences. We propose as the most promising approach to focus on the role of spokes in the process of electron energization.

In the search for a theoretical explanation of the CIV effect, a complication has been that it has been reported for widely different discharge geometries and magnetic topologies, and over orders of magnitude in gas pressure, plasma density, magnetic field strength, and current density. Already Sherman,⁶ in the first review of the theory of CIV, pointed out that it is unlikely one single theory can explain all these experiments, and yet they can in some sense be condensed into the simple formula of Eq. (1). In this situation, much of the theoretical work in the CIV literature has concentrated on a generic issue: Alfvén's original assumption of a mechanism for energy transfer from newly ionized neutrals to electrons. Analytical treatment,³⁴ weak turbulence theory,³⁵ particle in cell simulations,^{36,37} and experiments³⁸ all indicate that cross-**B** two-stream instabilities in the lower hybrid range of frequencies are likely to fill the role. Oscillations in this range of frequencies have also been reported from HiPIMS discharges and proposed to be involved in the electron transport mechanism.³⁹ In the theory for spokes in CIV discharges,⁴⁰ such instabilities are also proposed to play a central role.

Also in the HiPIMS discharge literature, the electron energization is a central issue. The proposed mechanisms can be separated into three categories: (1) acceleration of secondary-emitted electrons across the sheath, (2) acceleration in the sheath of electron that have been produced by electron impact inside it, and (3) Ohmic heating $\mathbf{J} \cdot \mathbf{E}$ by currents, carried by electrons, in the plasma outside the sheath. Note that the latter category contains both acceleration along the magnetic field (physically analogous to the mechanism of acceleration in the sheath), and electron currents carried across the magnetic field, which is fundamentally different. In Thornton,⁴¹ further developed by Depla *et al.*,⁴² categories (1) and (2) were analyzed and shown to lead to the Thornton equation, a condition on the minimum discharge voltage $U_{D,\min}$ as function of parameters that represent the probabilities of the included processes: secondary electron emission γ_e and recapture r , ionization in the sheath m , loss of energetic electrons out of the system ϵ_e , and the return probability β of the ions to the target. Later, model calculations,^{43,44} benchmarked against data from real HiPIMS discharges, showed that they generally operate at discharge voltages below that predicted by the Thornton equation. The proposed explanation was that the electron heating category (3) above should be added: an extended potential drop in the plasma outside the sheath, large enough to give significant Ohmic heating.

Energization of electrons inside the spokes, as proposed for CIV discharges,⁴³ belong to the category (3) of electron heating. Important questions here are how the spokes do arise, what effects they have on the current conduction and the electron energy balance, and how they influence the motion and energy spectrum of the ions. For these processes, there is a considerable amount of experimental observations, and theoretical analysis, in the CIV literature. A combination of the CIV and HiPIMS data sets, and theories, on spokes is the topic of a forthcoming publication.

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